DATABASE TECHNOLOGIES – ASSIGNMENT 8

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
SELECT *
FROM orders o
WHERE o.cnum = (SELECT cnum
                FROM customers
                WHERE cname = 'Cisneros');
```

```
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders o
    -> WHERE o.cnum = (SELECT cnum
    ->                 FROM customers
    ->                 WHERE cname = 'Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.163 sec)

[d5_96525_Deven@sales]>
```

7. Extract all the orders of Motika.

```
SELECT *
FROM orders o
WHERE o.snum = (SELECT snum
                FROM salespeople
                WHERE sname = 'Motika');
```

```
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders o
    -> WHERE o.snum = (SELECT snum
    ->          FROM salespeople
    ->          WHERE sname = 'Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.006 sec)

[d5_96525_Deven@sales]>
```

8. Find all the orders of salespeople servicing customers in London.

SELECT *

FROM orders o

WHERE o.cnum IN (SELECT cnum

FROM customers

WHERE city = 'London');

```
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders o
    -> WHERE o.cnum IN (SELECT cnum
    ->          FROM customers
    ->          WHERE city = 'London');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.123 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
SELECT *
FROM orders
WHERE amt > ANY (
   SELECT amt
   FROM orders
   WHERE cnum IN (
      SELECT cnum
      FROM customers
      WHERE city = 'London'
   )
);
```

```
                  FROM ' at line 4
[d5_96525_Deven@sales]>
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders
    -> WHERE amt > ANY (
    ->     SELECT amt
    ->     FROM orders
    ->     WHERE cnum IN (
    ->         SELECT cnum
    ->         FROM customers
    ->         WHERE city = 'London'
    ->     )
    -> );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.154 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
SELECT *

FROM customers c

WHERE c.rating >= ANY(SELECT rating

                      FROM customers

                      WHERE snum = (  SELECT snum

                                      FROM salespeople

                                      WHERE sname = 'serres'));
```

```
[d5_96525_Deven@sales]>SELECT *
    -> FROM customers c
    -> WHERE c.rating >= ANY(   SELECT rating
    ->                          FROM customers
    ->                          WHERE snum = (  SELECT snum
    ->                                          FROM salespeople
    ->                                          WHERE sname = 'serres'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.006 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
SELECT *

FROM customers

WHERE rating > ANY (

   SELECT rating

   FROM customers

   WHERE city = 'Rome'

);
```

```
[d5_96525_Deven@sales]>
[d5_96525_Deven@sales]>SELECT *
    -> FROM customers
    -> WHERE rating > ANY (
    ->     SELECT rating
    ->     FROM customers
    ->     WHERE city = 'Rome'
    -> );
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.008 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

SELECT *

FROM orders

WHERE amt > ANY(

SELECT amt

FROM orders

WHERE odate = '1990-10-04');

```
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders
    -> WHERE amt > ANY(
    -> SELECT amt
    -> FROM orders
    -> WHERE odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.005 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
SELECT *
FROM orders
WHERE amt < ANY(
        SELECT amt
        FROM orders
        WHERE cnum = ANY(
                SELECT cnum
                FROM customers
                WHERE city = 'san Jose'
                )
```

```
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders
    -> WHERE amt < ANY(
    ->  SELECT amt
    ->  FROM orders
    ->  WHERE cnum = ANY(
    ->                  SELECT cnum
    ->                  FROM customers
    ->                  WHERE city = 'san Jose'
    ->                  )
    ->          );
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.008 sec)
```

);

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

SELECT o.amt, c.cname, c.rating

FROM orders o, customers c

WHERE o.cnum = c.cnum

AND o.amt > (SELECT AVG(amt) FROM orders);

```
ERROR 1222 (21000): The used SELECT statements have a different number of columns
[d5_96525_Deven@sales]>
[d5_96525_Deven@sales]>SELECT o.amt, c.cname, c.rating
    -> FROM orders o, customers c
    -> WHERE o.cnum = c.cnum
    -> AND o.amt > (SELECT AVG(amt) FROM orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.014 sec)
```

14. Select those customers whose rating are higher than every customer in London.

SELECT *

FROM customers

WHERE rating > ANY (

SELECT rating

FROM customers

WHERE city = 'LONDON'

);

```
[d5_96525_Deven@sales]>
[d5_96525_Deven@sales]>SELECT *
    -> FROM customers
    -> WHERE rating > ANY (
    ->     SELECT rating
    ->     FROM customers
    ->     WHERE city = 'LONDON'
    -> );
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.009 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

SELECT sname, snum, city

FROM salespeople s

WHERE city != ALL (

SELECT city

FROM customers c

WHERE c.snum = s.snum

);

```
[d5_96525_Deven@sales]>SELECT sname, snum, city
    -> FROM salespeople s
    -> WHERE city != ALL (
    ->     SELECT city
    ->     FROM customers c
    ->     WHERE c.snum = s.snum
    -> );
+---------+------+-----------+
| sname   | snum | city      |
+---------+------+-----------+
| Motika  | 1004 | London    |
| Rifkin  | 1007 | Barcelona |
| Axelrod | 1003 | New York  |
+---------+------+-----------+
3 rows in set (0.011 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
SELECT sname, snum
FROM salespeople
WHERE snum = ANY (
    SELECT snum
    FROM customers
    GROUP BY snum
    HAVING COUNT(cnum) > 1
);
```

```
[d5_96525_Deven@sales]>SELECT sname, snum
    -> FROM salespeople
    -> WHERE snum = ANY (
    ->     SELECT snum
    ->     FROM customers
    ->     GROUP BY snum
    ->     HAVING COUNT(cnum) > 1
    -> );
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.124 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
SELECT
    s.snum,
    s.sname,
    s.city,
    (SELECT COUNT(c.cnum) FROM customers c WHERE c.snum = s.snum) AS customer_count
FROM salespeople s
WHERE s.snum IN (
```

SELECT snum

FROM customers

GROUP BY snum

HAVING COUNT(cnum) > 1

);

```
[d5_96525_Deven@sales]>SELECT
    ->     s.snum,
    ->     s.sname,
    ->     s.city,
    ->     (SELECT COUNT(c.cnum) FROM customers c WHERE c.snum = s.snum) AS customer_count
    -> FROM salespeople s
    -> WHERE s.snum IN (
    ->     SELECT snum
    ->     FROM customers
    ->     GROUP BY snum
    ->     HAVING COUNT(cnum) > 1
    -> );
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.010 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

SELECT snum,sum(amt)

FROM orders

GROUP BY snum

HAVING sum(amt) > ANY (SELECT MAX(o.amt) FROM orders o);

```
[d5_96525_Deven@(none)]>use hr;
Database changed
[d5_96525_Deven@hr]>use sales;
Database changed
[d5_96525_Deven@sales]>SELECT snum,sum(amt)
    -> FROM orders
    -> GROUP BY snum
    -> HAVING sum(amt) > ANY (SELECT MAX(o.amt) FROM orders o);
+------+----------+
| snum | sum(amt) |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.413 sec)
```